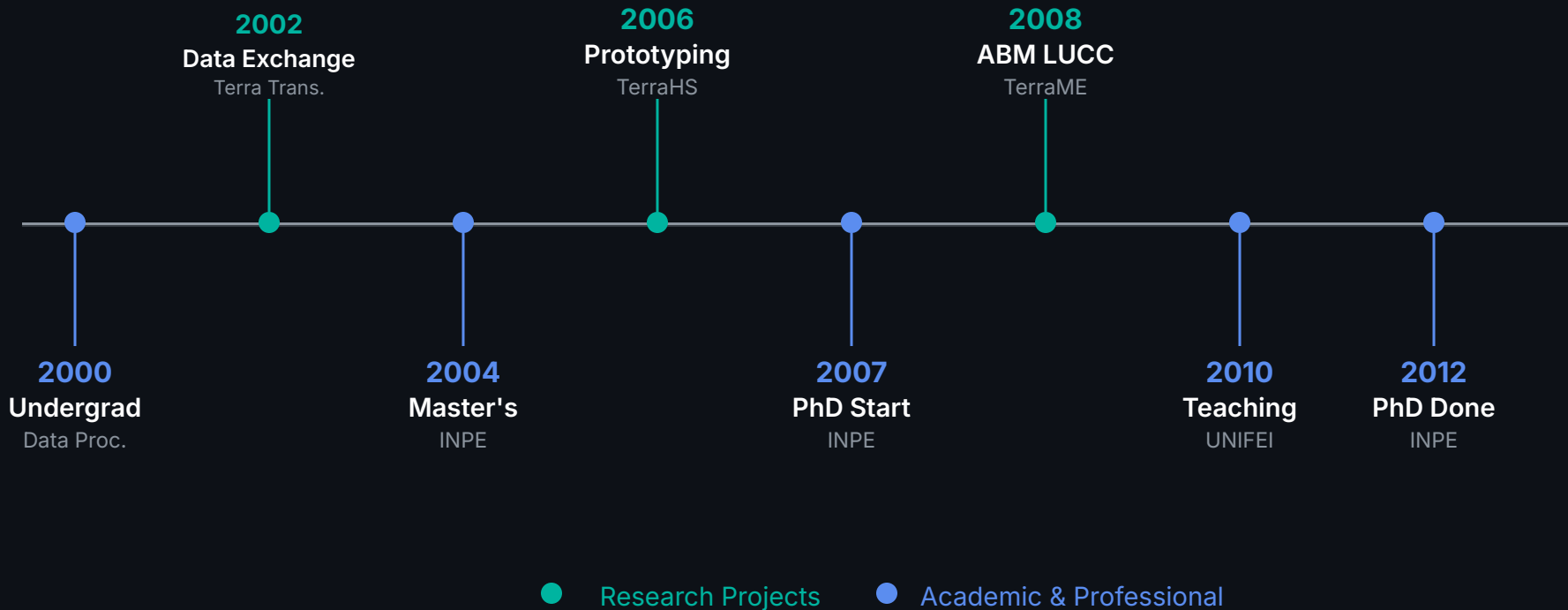


DisSModel

Building Open & Reproducible
Geospatial Simulations with Python

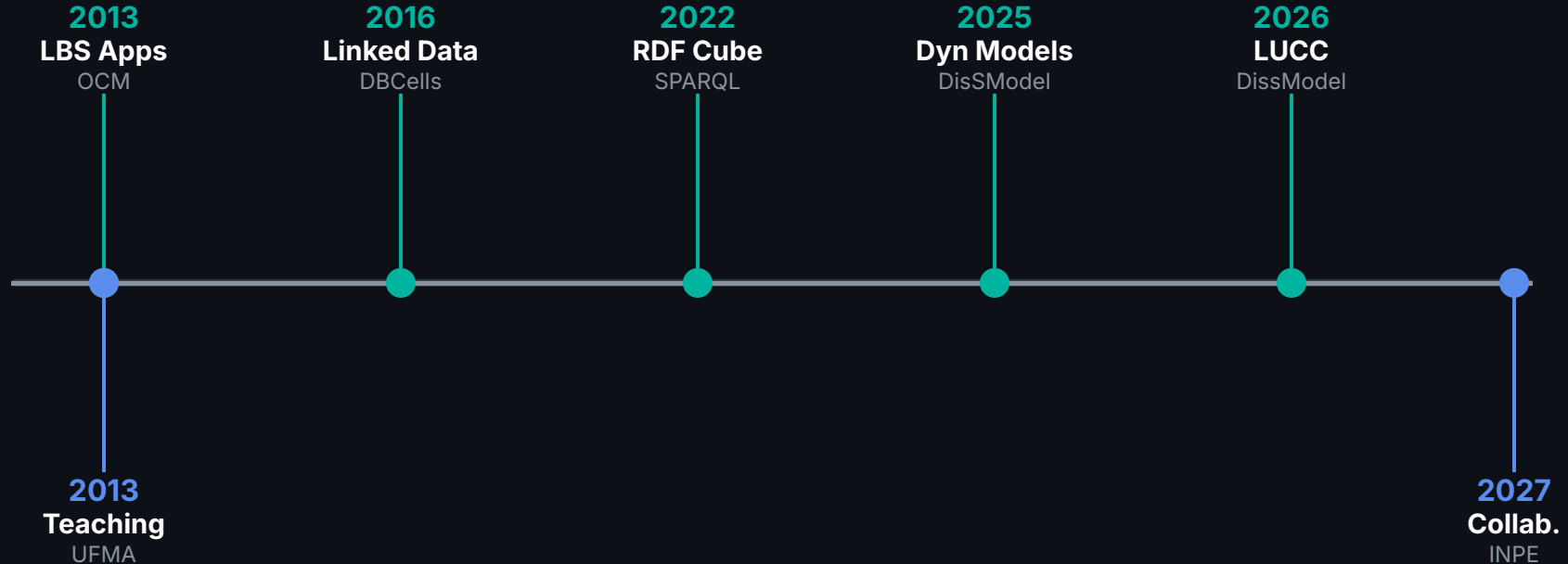
Career & Research

2000–2012 · From undergraduate to PhD — foundations in spatial modeling



Career & Research

2013–2027 · UFMA · Open Science · DisSModel · LuccME++



■ Research Projects

■ Academic & Professional

Science is not reproducible.

This is a documented crisis — not an opinion.

70%

of researchers cannot reproduce
experiments from other groups.

Baker (2016) · Nature

Three Root Failures



Code doesn't run

Implicit environments —
works only on author's
machine.



Data is missing

Input data not versioned. No
data = no audit.



Parameters unknown

Configs live in ad hoc scripts
or researcher's memory.

We model how territory changes.



Deforestation



**Urban
Expansion**



**Coastal
Dynamics**



**Agricultural
Frontier**

*Focus: **Process-based geospatial modeling** — not general-purpose ML.*

Existing tools are powerful...

...but they weren't built for reproducibility at scale.

Dinamica EGO

No support for process-based geospatial models (INPE)

TerraME

Last release 2020 · No cloud/STAC support

LuccME

Inherits Lua constraints · Weak Python link

DisSModel

Discrete Spatial Modeling Framework for Python

Open source · Reproducible · Cloud-native

```
pip install dissmode1
```

**"Science should not need
to be rewritten
to go into production."**

— DisSModel Principle

End-to-End Workflow

1

Create

Design model using base library

2

Package

Encapsulate in executors

3

Publish

Push to GitHub + catalog

4

Run

Same code: local or cloud

Core Pillars



Reproducibility

Executor + TOML
Docker isolation
SHA-256 audit trail



FAIR Software

Findable via Zenodo DOI
Accessible via REST API
Interoperable + Reusable



Layer Separation

Science: pure logic
Platform: APIs & infra
SimOps: operations

Three Components



Library **dissmodel**

- Core simulation engine.
- Local dev & execution.
- `pip install dissmodel`



Platform **dissmodel-platform**

- Cloud-native infra.
- FastAPI + Docker + Redis.
- `docker compose up`



Catalog **dissmodel-configs**

- Single source of truth.
- TOML manifests.
- Version-controlled PRs

Same code. Runs locally and on the remote platform.

Validation through laboratory models.

Cellular Automata · System Dynamics

Game of Life

Cellular Automaton · Built on GeoDataFrames

● ● ● game_of_life.py

```
class GameOfLife(CellularAutomaton):  
    def rule(self, idx) -> int:  
        state = self.gdf.loc[idx, 'state']  
        n = self.neighbor_values(idx).sum()  
  
        if state == 1:  
            return 1 if 2 <= n <= 3 else 0  
        return 1 if n == 3 else 0
```

```
# Environment Initialization  
gdf = vector_grid((20, 20), attrs={"state": 0})  
env = Environment()  
GameOfLife(gdf).initialize()
```

Parameters

Model

FireModelProb

Anneal

FireModel

FireModelProb

GameOfLife

Growth

Propagation

Snow

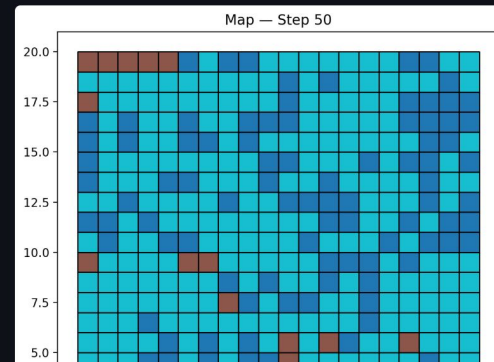
FireModelProb parameters

prob_combustion

0.00

prob_regrowth

Deploy



Cellular Automata Simulation

Visualization of grid state at step 50. The model uses a **Queen** neighborhood (8 neighbors) and classic Conway's rules.

SIR Model

System Dynamics · Flows, Stocks, and Feedback Loops

● ● ● sir_model.py

```
class SIR:
    def execute(self):
        total = S + I + R
        alpha = contacts * prob
        new_inf = I * alpha * (S/total)
        new_rec = I / duration

        S -= new_inf
        I += new_inf - new_rec
        R += new_rec
```

```
env = Environment()
SIR(9998, 2, 0, 2, 6, 0.25)
Chart(title="SIR Model")
env.run(30)
```

Parameters

Model

SIR

Coffee

Lorenz

PopulationGrowth

PredatorPrey

SIR

susceptible

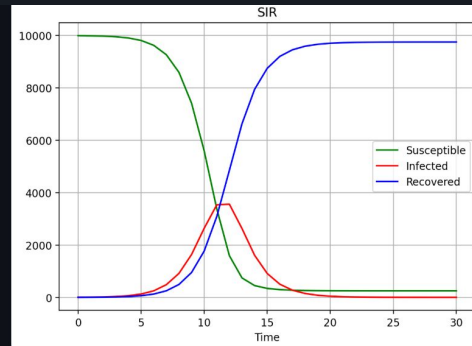
9998

infected

2

recovered

9



Dynamic Simulation Output

Visualization of Susceptible (green), Infected (red), and Recovered (blue) populations over 30 time steps. Parameters control the interaction rates and duration.

**Now: real-world
empirical models.**

From theory to application.

brmangue

Coastal dynamics model · Maranhão Coast, Brazil



FloodModel

Simulates flooding due to sea level rise scenarios based on terrain elevation and elevation rate (mm/yr).



Study Area

Ilha do Maranhão · Vector spatial grid · 97,309 cells
Scientific basis: Bezerra et al. (2014)



MangroveModel

Models mangrove migration driven by tidal height and soil type dynamics over time.



Key Parameters

Elevation rate · Tide height · Active accretion
Simulation: 2008 → 2030 (88 steps)
CRS: EPSG:31984 · Resolution: 100m

Coastal Case Study: brmangue

Mangrove migration + flooding · Maranhão Coast, Brazil

39.4×

Raster speedup
vs Vector

99.8%

Accuracy match
vs TerraME

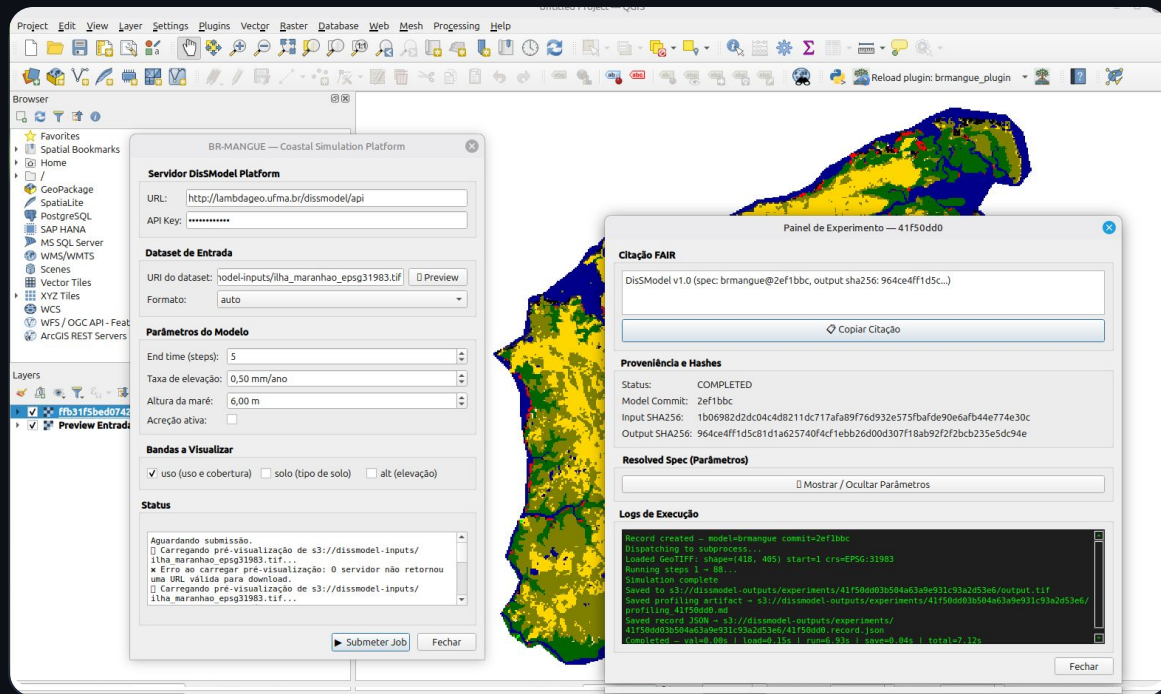
97K

cells at scale

QGIS Plugin available — simulation inside the GIS environment, no API calls required.

Coastal Case Study: brmangue

Plugin Integration & Visual Interface



Plugin Features

Integrates **brmangue** model for coastal dynamics simulation and visualization.

- Intuitive parameter input
- Direct output visualization
- Seamless vector grid integration

QGIS Plugin interface: Simulation executed directly within the GIS environment.

DisSLUCC

Land Use & Cover Change — Continuous CLUE-type allocation

1

Demand

Magnitude of change per timestep.
Pre-computed via historical CSV
data (MapBiomas/LuccME).

```
DemandPreComputedValues(  
    annual_demand=load_csv(...)  
)
```

2

Potential

Change suitability via linear
regression with spatial driver
variables.

```
PotentialLinearRegression(  
    gdf=gdf, drivers=[...]  
)
```

3

Allocation

CLUE-like distribution: cells receive
fractional proportions, not discrete
classes.

```
AllocationClueLike(  
    demand, potential  
)
```

DisSLUCC — Land Use & Cover Change

CLUE-type allocation · LuccME AC dataset (2008–2014)



Performance Speedup

3.9x

Raster speedup vs Vector

31.6 ms vs 122.5 ms per step



Accuracy Metrics

MAE = 0.002276

Accuracy vs TerraME

- Vector = Raster on both substrates
- Preserves full algorithmic correctness

Evaluation based on Willmott (2005) methodology

What Do We Gain?



Hidden Engineering

Data orchestration runs seamlessly — infrastructure is invisible to the researcher.



Full Traceability

SHA-256 hashes + FAIR citations guarantee provenance at every simulation stage.



Direct GIS Delivery

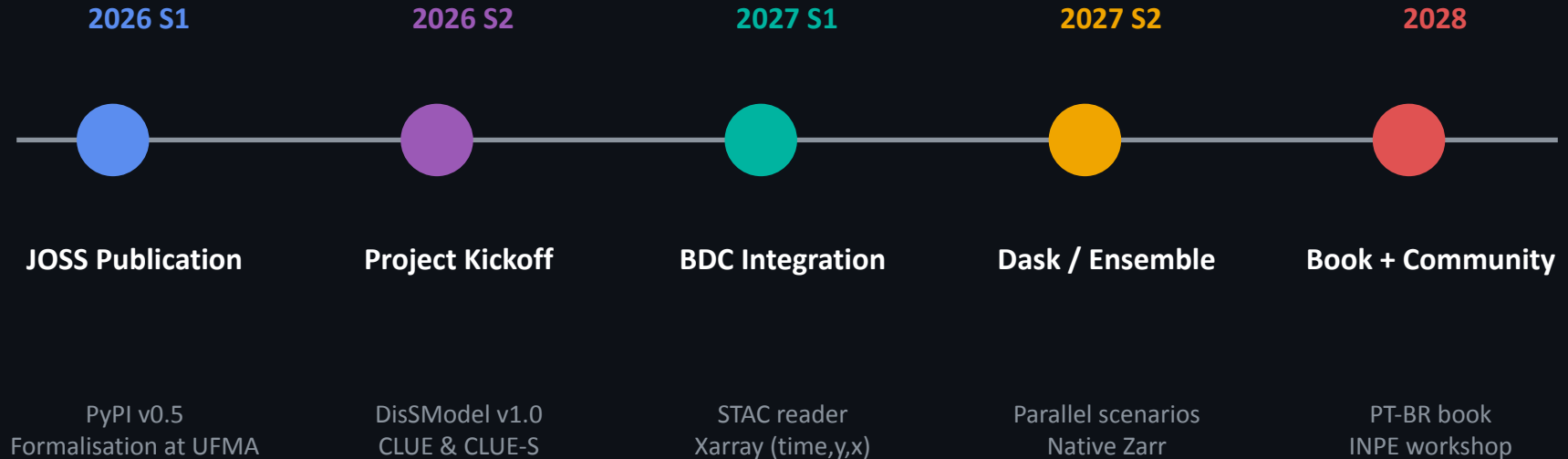
Results rendered natively in QGIS — no middleware, no extra steps.



End-to-End Focus

Researchers stay focused on science. Users stay focused on decisions.

Roadmap 2026–2028



Technology solves the how.

Governance solves the who and the where.

Infrastructure

Who maintains the platform? Server, storage, uptime.

Data Policy

Where do results live? Integration with Brazil Data Cube?

Model Catalog

Who reviews PRs? Acceptance criteria?

Community

Without shared infra, reproducibility problems remain.

DisSModel → LuccME++



2027/28 — DisSModel as the innovation laboratory for LuccME at CCST/INPE.

TerraME was the starting point — DisSModel advances as the foundation.

**Reproducibility
is not optional.**

It is the foundation of science.

Thank you!

DisSModel — open source · open science

GitHub

`disssmodel.github.io`

Docs

`disssmodel.github.io/disssmodel/`

Install

`pip install disssmodel`

Contact

`sergio.costa@ufma.br`

"Science should not need to be rewritten to go into production."

— DisSModel Principle